Name
Date
Course

Solving Quadratic Equations by Factoring

Solve the following quadratic equations.

① $x^2+6x+8=0$

$(x+4)(x+2)=0$

$x+4=0$ | $x+2=0$

$\boxed{x=-4}$ | $\boxed{x=-2}$

② $y^2+3y-28=0$

$(y+7)(y-4)=0$

$y+7=0$ | $y-4=0$

$\boxed{y=-7}$ | $\boxed{y=4}$

③ $z^2-9z+18=0$

$(z-6)(z-3)=0$

$z-6=0$ | $z-3=0$

$\boxed{z=6}$ | $\boxed{z=3}$

④ $y^2+2y+1=0$

$(y+1)(y+1)=0$

$y+1=0$ | $y+1=0$

$\boxed{y=-1}$ | $\boxed{y=-1}$

⑤ $x^2-4x-5=0$

$(x-5)(x+1)=0$

$x-5=0$ | $x+1=0$

$\boxed{x=5}$ | $\boxed{x=-1}$

⑥ $z^2-5z-24=0$

$(z+3)(z-8)=0$

$z+3=0$ | $z-8=0$

$\boxed{z=-3}$ | $\boxed{z=8}$

⑦ $x^2+12x+11=0$

$(x+11)(x+1)=0$

$x+11=0$ | $x+1=0$

$\boxed{x=-11}$ | $\boxed{x=-1}$

⑧ $y^2-4y+4=0$

$(y-2)(y-2)=0$

$y-2=0$ | $y-2=0$

$\boxed{y=2}$ | $\boxed{y=2}$

⑨ $0=z^2-7z+10$

$0=(z-5)(z-2)$

$z-5=0$ | $z-2=0$

$\boxed{z=5}$ | $\boxed{z=2}$

⑩ $0=x^2-3x+2$

$0=(x-2)(x-1)$

$x-2=0$ | $x-1=0$

$\boxed{x=2}$ | $\boxed{x=1}$

⑪ $0=y^2-6y-16$

$0=(y-8)(y+2)$

$y-8=0$ | $y+2=0$

$\boxed{y=8}$ | $\boxed{y=-2}$

⑫ $0=z^2-2z-35$

$0=(z-7)(z+5)$

$z-7=0$ | $z+5=0$

$\boxed{z=7}$ | $\boxed{z=-5}$

⑬ $2x^2+7x+3=0$

$(2x+1)(x+3)=0$

$2x+1=0$ | $x+3=0$

$2x=-1$ | $\boxed{x=-3}$

$\boxed{x=-\frac{1}{2}}$

⑭ $3y^2+y-4=0$

$(3y+4)(y-1)=0$

$3y+4=0$ | $y-1=0$

$3y=-4$ | $\boxed{y=1}$

$\boxed{y=-\frac{4}{3}}$

(15) $5z^2-6z+1=0$

$(5z-1)(z-1)=0$

$5z-1=0$ | $z-1=0$

$5z=1$ | $\boxed{z=1}$

$\boxed{z=\frac{1}{5}}$

(16) $2y^2+7y+6=0$

$(2y+3)(y+2)=0$

$2y+3=0$ | $y+2=0$

$2y=-3$ | $\boxed{y=-2}$

$\boxed{y=-\frac{3}{2}}$

(17) $3x^2-10x-8=0$

$(3x+2)(x-4)=0$

$3x+2=0$ | $x-4=0$

$3x=-2$ | $\boxed{x=4}$

$\boxed{x=-\frac{2}{3}}$

(18) $2y^2-7y-15=0$

$(2y+3)(y-5)=0$

$2y+3=0$ | $y-5=0$

$2y=-3$ | $\boxed{y=5}$

$\boxed{y=-\frac{3}{2}}$

(19) $3x^2-4x-4=0$

$(3x+2)(x-2)=0$

$3x+2=0$ | $x-2=0$

$3x=-2$ | $\boxed{x=2}$

$\boxed{x=-\frac{2}{3}}$

(20) $4x^2+11x+6=0$

$(4x+3)(x+2)=0$

$4x+3=0$ | $x+2=0$

$4x=-3$ | $\boxed{x=-2}$

$\boxed{x=-\frac{3}{4}}$

(21) $6y^2-13y-28=0$

$(2y-7)(3y+4)=0$

$2y-7=0$ | $3y+4=0$

$2y=7$ | $3y=-4$

$\boxed{y=\frac{7}{2}}$ | $\boxed{y=-\frac{4}{3}}$

(22) $10z^2-51z+27=0$

$(2z-9)(5z-3)=0$

$2z-9=0$ | $5z-3=0$

$2z=9$ | $5z=3$

$\boxed{z=\frac{9}{2}}$ | $\boxed{z=\frac{3}{5}}$

(23) $4x^2-8x-21=0$

$(2x-7)(2x+3)=0$

$2x-7=0$ | $2x+3=0$

$2x=7$ | $2x=-3$

$\boxed{x=\frac{7}{2}}$ | $\boxed{x=-\frac{3}{2}}$

(24) $15x^2+13x-6=0$

$(3x-1)(5x+6)=0$

$3x-1=0$ | $5x+6=0$

$3x=1$ | $5x=-6$

$\boxed{x=\frac{1}{3}}$ | $\boxed{x=\frac{-6}{5}}$

(25) $9y^2-12y+4=0$

$(3y-2)(3y-2)=0$

$3y-2=0$ | $3y-2=0$

$3y=2$ | $3y=2$

$\boxed{y=\frac{2}{3}}$ | $\boxed{y=\frac{2}{3}}$

(26) $4z^2+28z+49=0$

$(2z+7)(2z+7)=0$

$2z+7=0$ | $2z+7=0$

$2z=-7$

$\boxed{z=\frac{-7}{2}}$ | $\boxed{z=\frac{-7}{2}}$

Write the following quadratic equations in standard form; then solve.

(27) $x^2+9+6x=0$

$x^2+6x+9=0$

$(x+3)(x+3)=0$

$x+3=0$ | $x+3=0$

$\boxed{x=-3}$ | $\boxed{x=-3}$

(28) $-20+10x^2+17x=0$

$10x^2+17x-20=0$

$(2x+5)(5x-4)=0$

$2x+5=0$ | $5x-4=0$

$2x=-5$ | $5x=4$

$\boxed{x=-\frac{5}{2}}$ | $\boxed{x=\frac{4}{5}}$

(29) $10+7y+y^2=0$

$y^2+7y+10=0$

$(y+2)(y+5)=0$

$y+2=0$ | $y+5=0$

$\boxed{y=-2}$ | $\boxed{y=-5}$

(30) $25z+3z^2-18=0$

$3z^2+25z-18=0$

$(3z-2)(z+9)=0$

$3z-2=0$ | $z+9=0$

$3z=2$ | $\boxed{z=-9}$

$\boxed{z=\frac{2}{3}}$

(31) $x^2+9x=-8$

$x^2+9x+8=0$

$(x+1)(x+8)=0$

$x+1=0$ | $x+8=0$

$\boxed{x=-1}$ | $\boxed{x=-8}$

(32) $15=-8x-x^2$

$x^2+8x+15=0$

$(x+5)(x+3)=0$

$x+5=0$ | $x+3=0$

$\boxed{x=-5}$ | $\boxed{x=-3}$

(33) $-z^2=5z-14$

$0=z^2+5z-14$

$0=(z+7)(z-2)$

$z+7=0$ | $z-2=0$

$\boxed{z=-7}$ | $\boxed{z=2}$

(34) $5y-6=y^2$

$0=y^2-5y+6$

$0=(y-2)(y-3)$

$y-2=0$ | $y-3=0$

$\boxed{y=2}$ | $\boxed{y=3}$

(35) $3x^2+24=17x$

$3x^2-17x+24=0$

$(3x-8)(x-3)=0$

$3x-8=0$ | $x-3=0$

$3x=8$ | $\boxed{x=3}$

$\boxed{x=\frac{8}{3}}$

(36) $7y+6y^2=5$

$6y^2+7y=5$

$6y^2+7y-5=0$

$(2y-1)(3y+5)=0$

$2y-1=0$ | $3y+5=0$

$2y=1$ | $3y=-5$

$\boxed{y=\frac{1}{2}}$ | $\boxed{y=-\frac{5}{3}}$

(37) $-12x-7=-4x^2$

$4x^2-12x-7=0$

$(2x+1)(2x-7)=0$

$2x+1=0$ | $2x-7=0$

$2x=-1$ | $2x=7$

$\boxed{x=-\frac{1}{2}}$ | $\boxed{x=\frac{7}{2}}$

(38) $-7-6y=-y^2$

$y^2-6y-7=0$

$(y-7)(y+1)=0$

$y-7=0$ | $y+1=0$

$\boxed{y=7}$ | $\boxed{y=-1}$

39 $-100-z^2=20z$

$0=z^2+20z+100$

$0=(z+10)(z+10)$

$z+10=0$ | $z+10=0$

$\boxed{z=-10}$ | $\boxed{z=-10}$